Algebra 1 Name____________________
Unit 1 Date_____________________
Lesson 4 Homework Period___________________

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $(4.5^2)^5 =$

2. $\left(\frac{3}{7}\right)^{-3} =$

3. $(-2 \cdot 6)^4 =$

4. $\frac{(-3.1)^{10}}{(-3.1)^3} =$

5. $7^{-4} \cdot 7 \cdot 7^3 =$

Find the value of each for questions 6-7.

6. $\sqrt[3]{64} =$

7. $\sqrt{64} =$

8. Explain why the value of the $\sqrt{64}$ is different from the value of the $\sqrt[3]{64}$.

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{y} \cdot \sqrt{y} = 15$

$y =$ ______

10. $\sqrt[3]{g} \cdot \sqrt[3]{g} \cdot \sqrt[3]{g} = -3$

$g =$ ______